

The Education Debugger

An AI-powered Smart Education platform that finds why a student is struggling, repairs the underlying gap, and adapts learning to the student.

Core loop	Learn -> Practice -> Teach-back -> Diagnose -> Repair -> Adapt
SIH direction	Student Innovation - Smart Education
Prototype scope	One subject, 20-30 concepts, student + teacher interfaces, simulated/seeded data, AI-assisted diagnosis.

Prototype principle: Don't just show what a student got wrong. Show what caused it and what to do next.

1. The Problem We Are Solving

Most learning systems are good at recording answers and scores, but weak at explaining the underlying reason for a learning failure. A student can repeatedly fail a topic because an earlier prerequisite is weak, a misconception is stable, or the chosen learning method is ineffective for that student.

Observed symptom	Low marks / repeated mistakes / slow progress
What existing systems often do	Recommend more content on the same topic
What our system asks	What concept, misconception, prerequisite, or learning mode is actually causing the problem?
Outcome	A targeted repair path instead of generic extra study

Design north star

Education should behave like a good debugger. When something fails, first locate the root cause, then repair it, then rerun the test.

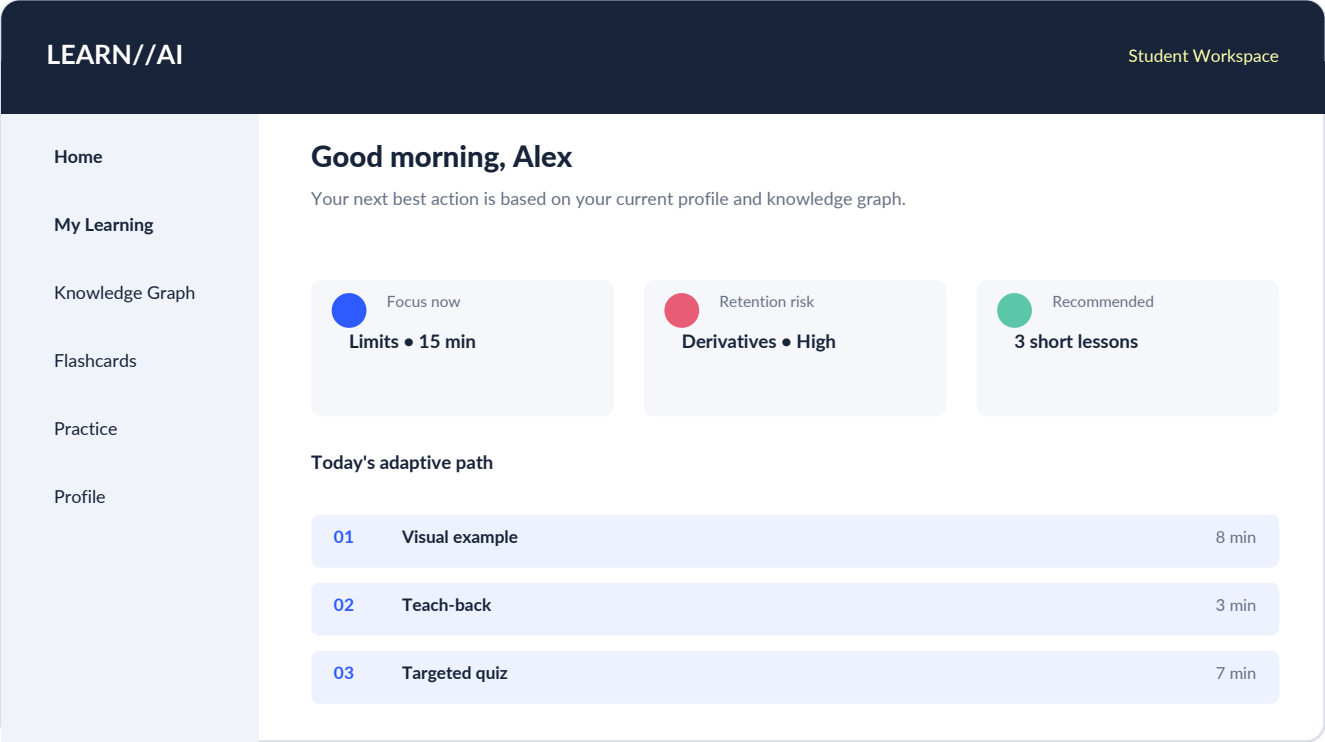
2. Product Concept

LEARN//AI combines the strongest ideas you listed into one coherent product. The center is a dynamic Knowledge Graph; around it sit Reverse Tutoring, Misconception Genome, Prerequisite X-Ray, Personal Learning Profile, Adaptive Difficulty, Flashcards / Retrieval, and a Don't Study This Yet prioritization engine.

Module	What it does
Prerequisite X-Ray	Traces a struggling topic backward to likely prerequisite gaps.
Reverse Tutoring	Student explains the concept; AI evaluates depth, missing pieces and misconceptions.
Misconception Genome	Learns recurring patterns in the student's wrong reasoning, not only wrong answers.
Knowledge Graph	Maps concepts and prerequisite relationships; supports root-cause diagnosis.
Personal Learning Profile	Learns which formats and study methods produce better retention for the student.
Adaptive Difficulty	Changes question complexity using performance, speed, hints and confidence signals.
Flashcards + Retrieval	Schedules short revision around concepts at risk of decay.
Don't Study This Yet	Prioritizes topics using current mastery, prerequisites, exam relevance and time available.
Education Time Machine	Looks backward through prior learning history to find when a gap likely began.
Class Knowledge Map	Aggregates concept mastery across a cohort so teachers see class bottlenecks.

3. Student Interface - Dashboard

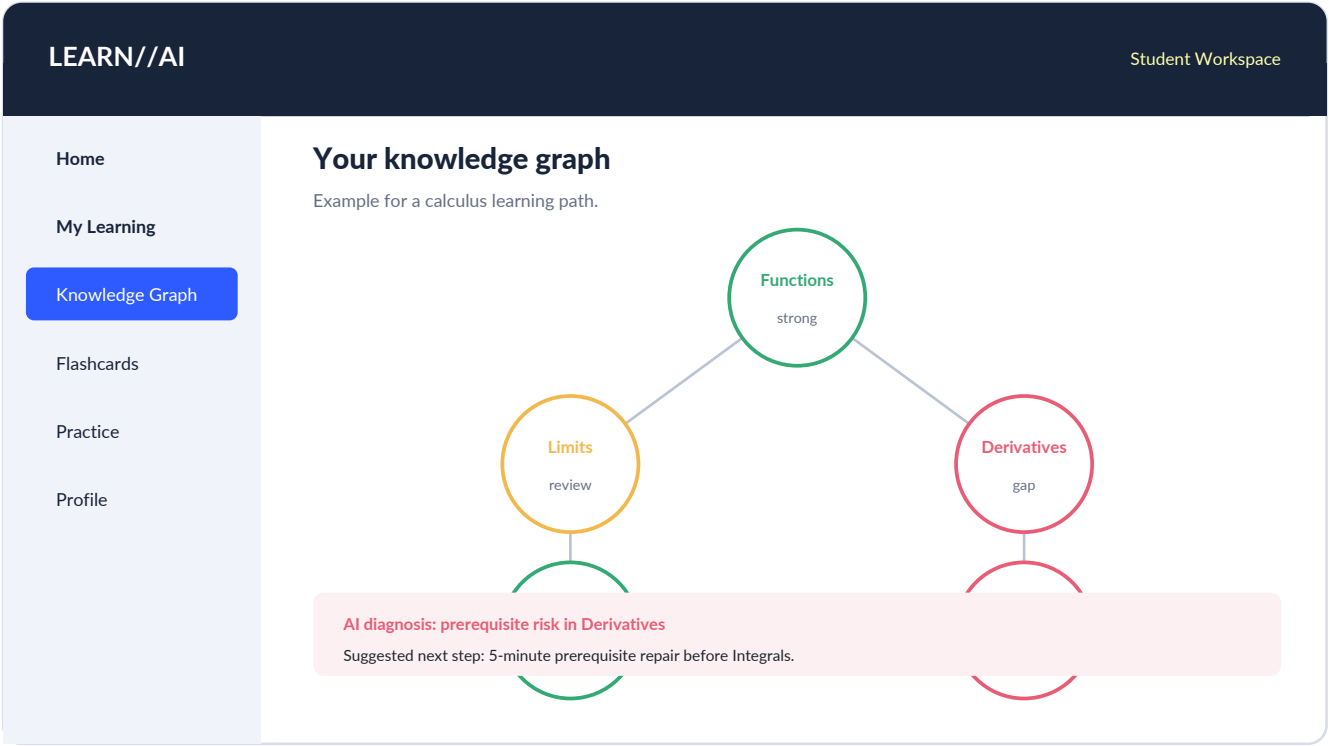
The dashboard should answer three questions immediately: What should I study now? Why? What is the shortest useful next action?



Key UI rule: Never overwhelm the student with 20 recommendations. Surface one high-value next step, then let them explore.

4. Student Interface - Knowledge Graph + Prerequisite X-Ray

The knowledge graph is the product's main reasoning surface. Green means strong, amber means needs review, red means likely gap. The AI can trace a red node backward through dependencies.



Example diagnosis:

Current struggle: Integrals. Likely prerequisite chain: Integrals -> Derivatives -> Limits -> Functions. If the evidence is strongest around Derivatives, the platform recommends repairing Derivatives before pushing more Integrals content.

5. Reverse Tutoring - The Signature Interaction

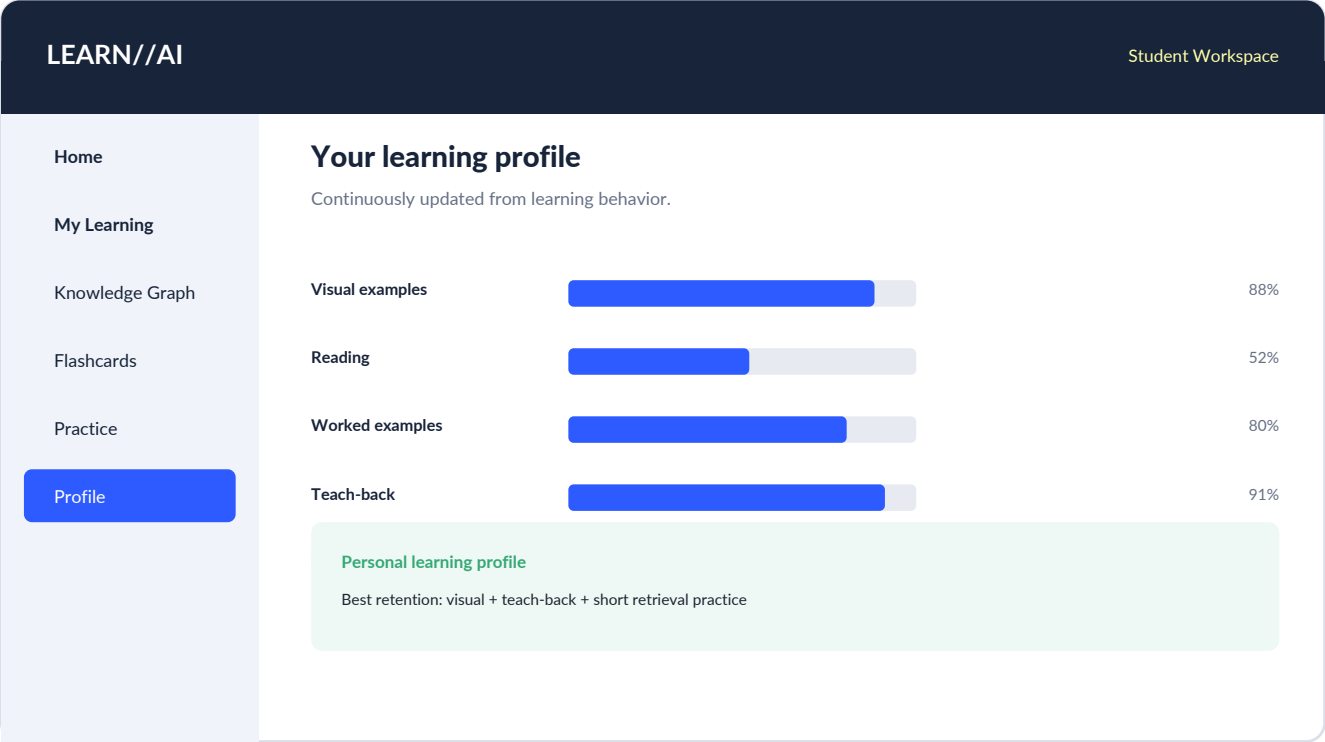
After a lesson, the student is asked to teach the concept back in their own words. The system does not need to judge presentation style; it evaluates whether important concepts are present, connected and internally consistent.

Student prompt	Explain "why the derivative is useful" as if you were teaching a friend.
AI checks	Concept coverage, prerequisite understanding, contradictions, missing links, confidence signals, and repeated misconception patterns.
Output	Understood: rate of change. Missing: connection to slope. Review: limits. Next action: 3-minute visual example + one teach-back.

MVP note: In the first prototype, the teach-back input can be typed text. Voice/video analysis can be a later extension once your team has reliable data and evaluation methods.

6. Personal Learning Profile

The profile is built from behavior, not a one-time personality quiz. It tracks which study formats produce better short-term performance and longer-term retention.



Signal	Use
Accuracy	Estimate mastery per concept.
Time / hints	Detect fluency vs. guessing.
Retrieval after delay	Estimate retention / decay risk.
Format performance	Choose examples, reading, flashcards, teach-back, etc.

7. Adaptive Difficulty + "Don't Study This Yet" Engine

The system should distinguish importance from urgency. A difficult topic is not always the best next topic. The best next action depends on prerequisites, mastery, exam relevance and available time.

Student state	System decision
High importance + weak prerequisite	Study now: repair prerequisite first.
Low importance + weak topic	Defer: don't study this yet.
High mastery + high decay risk	Refresh: short flashcard/retrieval session.
High accuracy + low confidence	Calibrate: challenge slightly harder and collect confidence.
Low accuracy + many hints	Repair: return to prerequisite / worked example.

Example priority score Priority = topic importance x gap severity x prerequisite impact x time fit	Guardrail The system should explain the recommendation: "Study X first because it unlocks Y and is currently your largest dependency gap."
---	--

8. Teacher Interface - Upload to Teaching Plan

The teacher interface should reduce workload rather than create a second LMS. A teacher uploads existing material, reviews AI suggestions, and uses the resulting concept map and class insights.

LEARN//AI

Teacher Console

Overview

Upload lesson

Concept map

Class gaps

Students

Assignments

Upload a lesson

AI extracts concepts, prerequisites and draft questions for teacher review.

Drop PPT / PDF here

or choose from device

Analyze material

Teacher review required

Teacher control is essential: AI drafts; teacher approves, edits, or rejects. This is especially important for concept explanations, assessment questions, and prerequisite relationships.

Smart Education - AI Learning Debugger | SIH Internal Prototype

9

9. Teacher Interface - Class Knowledge Map

The teacher should not have to inspect 50 individual answer sheets to discover a common learning bottleneck. The dashboard aggregates results by concept and misconception cluster.

LEARN//AI

Teacher Console

Overview

Upload lesson

Concept map

Class gaps

Students

Assignments

Class learning gaps

Identify shared misconceptions and prerequisite bottlenecks.

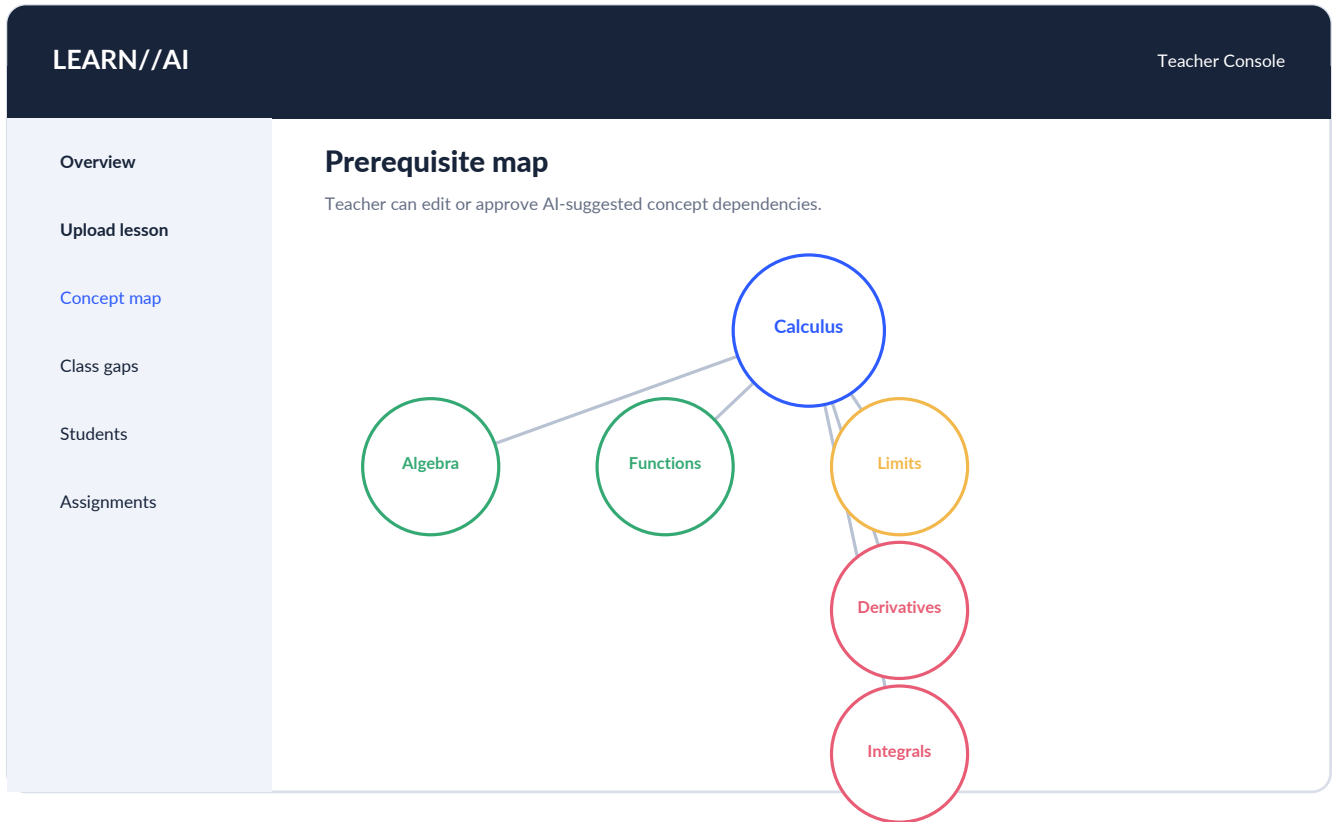
Concept	Mastery	Signal	Suggested action
Functions	91%	Strong	Continue
Limits	64%	Watch	Retrieval practice
Derivatives	38%	High gap	Re-teach prerequisite
Integrals	31%	High gap	Repair derivatives first

AI insight: 62% of the class shows the same derivative misconception.

Action-oriented output: "62% of the class has the same derivative misconception. Review the prerequisite and assign a 5-question diagnostic before moving to integrals."

10. Teacher Interface - Editable Prerequisite Map

AI can propose dependencies; teachers can edit them. This keeps curriculum knowledge transparent and correctable instead of turning the graph into an opaque black box.



Recommended controls: Add concept, Connect prerequisite, Approve AI edge, Flag uncertain edge, View evidence.

11. Misconception Genome + Education Time Machine

These two modules answer different questions. The Misconception Genome asks: "How is this student repeatedly getting things wrong?" The Education Time Machine asks: "When in the learning history did the dependency gap likely begin?"

Misconception Genome	Education Time Machine
Clusters errors into patterns such as formula confusion, concept reversal, prerequisite confusion, unit/notation issues, or guessing.	Links current performance to earlier assessments/lessons and traces backward through the knowledge graph.
Output example: "Repeatedly treats derivative as a value rather than a rate of change."	Output example: "Current derivative difficulty correlates with a weak limits checkpoint 4 weeks earlier."

This is where your project becomes more than an AI tutor: it becomes an AI diagnostic layer for learning itself.

12. AI + System Architecture

Keep the first version modular. Do not build a giant model. Use proven services for document understanding and explanation, and combine them with transparent rules/analytics for the learning engine.

Layer	Prototype choice	Purpose
Web UI	HTML/CSS/JS or React	Student + teacher workflows
Backend	Python + FastAPI/Flask	APIs and orchestration
Database	MySQL	Students, concepts, attempts, progress, graph edges
AI / LLM	LLM/API + Python	Concept extraction, question generation, teach-back analysis
Learning engine	Python rules + basic ML later	Mastery, decay risk, priority, adaptive difficulty
Knowledge graph	SQL tables first; graph DB later	Prerequisite relationships and evidence
Analytics	Python + simple charts	Class concept mastery and interventions

AI safety/design principle: use AI to assist diagnosis and generation, but expose evidence and give teachers/student users a way to correct wrong suggestions.

13. Data Model - What You Need to Store

Entity	Core fields
Student	id, grade/year, goals, consent/settings
Concept	id, subject, title, difficulty, metadata
PrerequisiteEdge	from_concept, to_concept, confidence, teacher_verified
AssessmentItem	question, concept, difficulty, type
Attempt	student, question, answer, correctness, time, hints, confidence
Mastery	student, concept, mastery_score, last_seen, retention_risk
Misconception	student, pattern_id, evidence, confidence
LearningMethodResult	student, method, concept, outcome
StudyAction	student, concept, action_type, timestamp, result

For the first demo, you can seed a small dataset rather than waiting for thousands of real users. Make the data model realistic even if the data volume is small.

14. What We Should Actually Build for the Internal SIH Round

The biggest mistake would be trying to implement every feature. Build a thin but complete vertical slice that demonstrates the unique reasoning loop.

Phase	Scope
MVP - must work	Student login/demo profile; teacher uploads one PPT/PDF; AI extracts 10-20 concepts; editable concept/prerequisite map; 10-question diagnostic; mastery dashboard; prerequisite gap alert; reverse tutoring using typed response; personalized next action; teacher class-gap view.
Strong demo extras	Flashcard queue; adaptive question difficulty; "Don't study this yet" priority panel; misconception cluster view; simple retention-risk score.
Future	Voice/video teach-back; advanced ML; graph database; large-scale curriculum coverage; mobile app; automated content alignment; richer long-term retention modeling.

The demo story	A student is failing Integrals. The system discovers the prerequisite problem is Derivatives, asks the student to teach back the concept, detects a misconception, repairs the gap, changes difficulty, and finally shows measurable improvement.
-----------------------	---

15. Team Split + 3-Day Sprint

Role	Primary responsibility
Member A	Frontend + student screens
Member B	Backend + SQL + APIs
Member C	AI integration + prompt/evaluation logic
Member D	Concept graph + analytics + seed data
Member E (if available)	Integration + PPT + demo story + testing

Suggested 3-day sprint

Day 1: Freeze problem statement, UX flow, concept graph, wireframes, seed data. Day 2: Build the vertical slice and connect AI. Day 3: Test the full demo, prepare PPT, create screenshots, rehearse judge questions.

16. SIH Judge Questions You Should Be Ready For

What is innovative?	We do not stop at score-based recommendation. We diagnose prerequisite gaps and misconception patterns, then adapt the learning path.
Why AI?	AI helps extract concepts, analyze open-ended teach-back, cluster misconceptions and personalize recommendations. Transparent rules handle core scoring logic.
What is different from an AI tutor?	An AI tutor answers questions. Our system maintains a structured learner model and diagnoses the reason behind learning failure.
How do you know a prerequisite is weak?	Use evidence from related assessment items, concept dependency edges, repeated errors and historical performance; show the evidence behind the recommendation.
Can this scale?	Yes. Start with one subject and a verified concept graph, then expand subjects, institutions and richer models.
What if AI is wrong?	Teacher approval, confidence scores, evidence views, and editable concept/relationship maps provide human oversight.
What is the measurable impact?	Pre/post mastery on targeted concepts, reduced repeated errors, improved delayed-retention performance, and teacher time saved in diagnosing class gaps.

One-line pitch	LEARN//AI is an AI-powered education debugger that finds the root cause of learning gaps, helps students teach concepts back, and continuously adapts what they should learn next.
-----------------------	--